Evolution in Audio Deepfake Detection

A Comparative Study of RawNet2, AASIST, AASIST3, Rawformer, and FAN-AASIST

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AI Ethics Disclosure

This report involves the use of Artificial Intelligence tools (I used ChatGPT and Claude.ai) for assistance in writing, and debugging only. Coding portion was manually done by me and all the codes were understood in depth by me. All outputs have been manually reviewed, validated, and adapted by me. I ensures that the work adheres to academic integrity guidelines, avoids plagiarism, and responsibly evaluates the societal impact of audio deepfake detection technologies.

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1. Abstract

Audio deepfake detection has emerged as a critical challenge in the era of rapidly advancing speech synthesis and voice conversion technologies. This report presents a systematic study of five deep learning architectures for audio anti-spoofing: RawNet2, AASIST, AASIST3, Rawformer, and a novel proposed variant, FAN-AASIST. All models are trained and evaluated on the ASVspoof 2019 Logical Access (LA) dataset under a unified experimental protocol using 2-second audio clips sampled at 16 kHz. The study traces the architectural evolution from 1D convolutional pipelines to spectro-temporal graph attention networks and transformer-based encoders, analysing each in terms of detection performance (EER, AUC, F1), parameter efficiency, computational cost (FLOPs/MACs), and memory footprint. Rawformer achieves the best overall detection performance (EER = 0.1427, AUC = 0.9426, F1-Macro = 0.7369), while AASIST remains the most parameter-efficient model at only 298K parameters. The proposed FAN-AASIST, which replaces graph attention with Fourier Analysis Network (FAN) layers, shows promise as a lightweight alternative but requires further tuning.

2. Introduction

Deepfakes refer to AI-generated or manipulated content produced via deep learning. In the audio domain, sophisticated text-to-speech (TTS) and voice conversion (VC) systems can now synthesise speech that is perceptually indistinguishable from genuine human voice. This capability poses serious threats to biometric authentication systems, voice-controlled assistants, and social media platforms where audio authenticity is assumed.

Audio Anti-Spoofing (AAS)—also known as audio deepfake detection—aims to build countermeasure (CM) systems that can classify an utterance as *bonafide* (genuine) or *spoof* (synthetic/converted). The ASVspoof challenge series has driven significant progress, providing standardised datasets and evaluation protocols.

This project surveys four leading architectures and proposes one new variant, studying their design philosophy, key innovations, empirical performance, and computational trade-offs. The central research question is: **how does the choice of sequence modelling strategy—recurrent, graph-attention, or transformer-based—affect detection accuracy and efficiency?**

2.1 Scope of Work

The scope of this project encompasses:

- Literature survey and architecture deep-dive of RawNet2, AASIST, AASIST3, and Rawformer.
- Implementation and training of all four models on the ASVspoof 2019 LA dataset.
- Design and experimentation with FAN-AASIST, a novel variant replacing graph attention with Fourier Analysis Network (FAN) layers.
- Comprehensive evaluation using EER, AUC, F1-Macro/Micro, Balanced Accuracy, Hamming Loss, and per-model computational profiling (MACs, FLOPs, GFLOPs/sec,

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memory usage).

- Analysis of training dynamics across 5 epochs and discussion of failure modes.

3. Problem Formulation

Let $\mathbf{x} \in \mathbb{R}^T$ denote a raw audio waveform of T samples. The binary classification task is:

$$f(\mathbf{x}) \longrightarrow \{\textit{bonafide},\ \textit{spoof}\} \tag{1}$$

Models are evaluated using the **Equal Error Rate (EER)**, which is the operating point at which the false acceptance rate (FAR) equals the false rejection rate (FRR). Lower EER indicates better performance. The Area Under the ROC Curve (AUC) and F1-score provide complementary evaluation under class imbalance.

The ASVspoof 2019 Logical Access dataset provides training, development, and evaluation partitions containing bonafide utterances from VCTK and spoofed utterances generated by 17 TTS/VC systems (attacks A01–A17). The class distribution is highly imbalanced, with spoofed utterances significantly outnumbering genuine ones.

4. Model Architectures

4.1 RawNet2 — End-to-End 1D Convolutional Pipeline

RawNet2 [2] processes raw waveforms directly using a Sinc-based convolutional front-end followed by residual CNN blocks with channel attention (FMS), and a GRU-based temporal aggregator.

4.1.1 Front-end: SincConv

The first layer applies 128 learnable sinc-function band-pass filters directly to the raw waveform. Each filter is parameterised by two cutoff frequencies (f_1, f_2), so the filter response is:

$$g[n] = 2f_2 \operatorname{sinc}(2\pi f_2 n) \ - \ 2f_1 \operatorname{sinc}(2\pi f_1 n) \tag{2}$$

Filters are initialised on the mel scale, providing a physically meaningful and parameter-efficient spectral decomposition. This avoids learning arbitrary convolution weights and ensures the front-end captures auditory-relevant frequency bands.

4.1.2 Residual Blocks with Feature Map Scaling (FMS)

Six residual blocks progressively expand channels (128 → 512) and compress the temporal dimension via max-pooling (stride 3), reducing 32,000 samples to 14 temporal frames. Each block is followed by a Feature Map Scaling (FMS) module—a squeeze-and-excitation style channel attention:

$$\mathbf{s} = \sigma\big(W_2 \cdot \mathrm{ReLU}(W_1 \cdot \mathrm{GAP}(\mathbf{x}))\big), \qquad \hat{\mathbf{x}} = \mathbf{x} \odot \mathbf{s} \tag{3}$$

where GAP denotes global average pooling. FMS allows the model to focus on discriminative frequency bands per utterance.

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4.1.3 GRU Temporal Aggregator

A GRU layer (hidden size 1024) processes the T=14 frame sequence, capturing long-range temporal dependencies. The final hidden state is passed to a 2-layer FC classifier.

Key insight: RawNet2 is the largest model at ∼12.8M parameters, dominated by its GRU (∼4.7M) and FC layers (∼1.1M). Its 1D pipeline preserves temporal resolution but at a substantial computational cost.

4.2 AASIST — Spectro-Temporal Graph Attention

AASIST [3] introduces a 2D spectro-temporal CNN front-end feeding into heterogeneous graph attention layers. At only ∼298K parameters, it is the most compact model in this study.

4.2.1 2D Spectro-Temporal Front-end

The SincConv output is reshaped into a 2D feature map (frequency × time), processed by 6 Conv2D residual blocks. The final feature map $\mathbf{F} \in \mathbb{R}^{64\times 23\times 14}$ is split to create two graph representations:

- **Spectral graph** $\mathcal{G}_s$: 23 nodes (frequency bins), each node is a 64-dim vector.
- **Temporal graph** $\mathcal{G}_t$: 14 nodes (time frames), each node is a 64-dim vector.

4.2.2 Graph Attention Layers (GAL)

Each graph undergoes graph attention, where node embeddings are updated by attending to neighbours. The attention coefficient between nodes i and j is:

$$\alpha_{ij} = \mathrm{softmax}_j\Big(\mathrm{LeakyReLU}\big(\mathbf{a}^\top[W\mathbf{h}_i \,\|\, W\mathbf{h}_j]\big)\Big) \tag{4}$$

Graph pooling reduces spectral nodes (23 → 11) and temporal nodes (14 → 9) before heterogeneous fusion.

4.2.3 Heterogeneous Graph Attention (HtrgGAL)

The key innovation of AASIST is modelling cross-modal interactions between spectral and temporal branches. The HtrgGAL module combines the two graphs and applies attention across node types, enabling the model to detect artifacts that span both frequency and time—a hallmark of modern TTS synthesis. The readout concatenates element-wise max-pooled vectors from both branches and passes them to a linear classifier.

4.3 AASIST3 — KAN-Enhanced Graph Attention

AASIST3 [4] replaces all standard Linear layers within the graph attention and pooling modules of AASIST with KANLinear layers from the Kolmogorov–Arnold Network (KAN) framework.

4.3.1 KANLinear Layers

Unlike a standard linear layer $\mathbf{y} = W\mathbf{x} + \mathbf{b}$, a KANLinear block pairs each weight with a learnable PReLU activation function:

$$y_j = \sum_i \varphi_{ij}(x_i), \qquad \text{where} \quad \varphi_{ij}(x) = \mathrm{PReLU}(w_{ij} \cdot x) \tag{5}$$

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This per-connection learnable non-linearity enables richer function approximation without structural changes, following the Kolmogorov–Arnold representation theorem.

4.3.2 Parameter Impact

Each KANLinear layer is approximately 14.6× larger than its Linear counterpart, because each weight now also carries a PReLU parameter. The CNN backbone is unchanged, so the overall parameter increase is ∼4.9× (298K → 1.45M). The structural topology of AASIST—spectral/temporal graphs, HtrgGAL, pooling, readout—remains identical.

4.4 Rawformer — Transformer Encoder with Positional Aggregation

Rawformer [5] replaces the graph back-end with a standard Transformer encoder, making it the only model in this study that uses multi-head self-attention (MHSA) in the traditional sense.

4.4.1 Architecture

- SincConv front-end (70 filters) on 4-second input (64,000 samples at 16 kHz).
- 4 Conv2D residual blocks producing a feature map of shape $64 \times 23 \times 16$.
- 2D Positional Encoding applied to the feature map before flattening.
- Flattening into 368 tokens of dimension 64.
- 2 Transformer encoder layers (MHSA + FFN, d=64), ∼102K parameters each.
- Sequence pooling: learned attention-weighted pooling over all 368 tokens → 64-dim vector.
- Linear classifier → single binary logit (sigmoid at inference).

4.4.2 Key Design Choice: Positional Aggregation

Unlike AASIST which explicitly separates spectral and temporal graphs, Rawformer preserves spatial structure implicitly through positional encoding. The longer 4-second input context provides richer temporal information. The Transformer's global self-attention allows every token to attend to every other, modelling long-range spectro-temporal dependencies.

4.5 FAN-AASIST — Proposed Novel Variant

FAN-AASIST is the novel contribution of this project. Motivated by the observation that deepfake audio artifacts often manifest as periodic inconsistencies in prosody, pitch, and formant structure, the hypothesis is that **Fourier Analysis Network (FAN) layers** may be better suited than self-attention for capturing such periodic patterns.

4.5.1 Fourier Analysis Network (FAN) Layer

A FAN layer [6] decomposes each input into periodic and non-periodic components, in contrast to self-attention which computes dense pairwise interactions:

$$\phi(\mathbf{x}) = \left[\cos(W_p\mathbf{x}) \parallel \sin(W_p\mathbf{x}) \parallel \sigma(\bar{B} + \bar{W}\mathbf{x})\right] \tag{6}$$

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The first two terms capture periodic structure via cosine and sine projections with learnable frequency matrix W_p, while the third term $\sigma(\bar{B} + \bar{W}\mathbf{x})$ provides non-periodic (standard MLP) components.

4.5.2 Integration into AASIST

In FAN-AASIST, the attention mechanism in HtrgGAL is replaced by FAN layers. All other sub-modules—RawNet2 encoder, spectral/temporal graph construction, graph pooling, element-wise max/min operator, and KAN-Linear readout—remain unchanged from AASIST. This yields a ∼136K parameter model (0.52 MiB), significantly smaller than AASIST.

4.5.3 Motivation

Self-attention computes $\mathcal{O}(N^2)$ pairwise interactions and may not naturally capture periodic artifacts. FAN, by explicitly encoding periodicity, could in principle detect phase inconsistencies introduced by vocoders or neural TTS systems—a lightweight and periodicity-aware alternative to standard attention.

5. Experimental Setup

Table 1: Experimental Configuration

Parameter	Value
Dataset	ASVspoof 2019 Logical Access (LA)
Audio Duration	2 seconds (zero-padded / truncated)
Sampling Rate	16,000 Hz
SincConv Filters	70
Encoder ResBlocks	6
Optimizer	Adam
Learning Rate	1×10^{-4}
Epochs	5
Loss Function	Binary / Softmax Cross-Entropy
Eval. Metrics	EER, AUC, F1 (Macro/Micro), Balanced Accuracy, Hamming Loss

All models receive input shape $[1, 32000]$, except Rawformer which uses $[1, 64000]$. No data augmentation was applied.

6. Experimental Results

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6.1 Detection Performance

Table 2: Detection performance metrics. Best results highlighted in **bold**.

Model	F1 Macro	F1 Micro	Prec. Macro	Recall Macro	AUC	EER
RawNet2	0.4482	0.4940	0.5841	0.7164	0.8732	0.2214
AASIST	0.5553	0.6466	0.6031	0.7750	0.9081	0.1821
AASIST3	0.5620	0.6542	0.6065	0.7833	0.9247	0.1640
Rawformer	**0.7369**	**0.8592**	**0.6971**	**0.8592**	**0.9426**	**0.1427**
FAN-AASIST	0.6147	0.7541	0.6101	0.7473	0.8275	0.2515

Rawformer achieves the best performance across all metrics: EER = 0.1427, AUC = 0.9426, and F1-Macro = 0.7369. AASIST3 improves upon AASIST (EER 0.164 vs. 0.1821), confirming that KAN layers provide a measurable benefit. RawNet2 underperforms despite its large parameter count, likely due to overfitting within only 5 training epochs. FAN-AASIST achieves a reasonable AUC (0.8275) but its EER (0.2515) is worse than all baselines.

6.2 Model Complexity and Computational Cost

Table 3: Computational complexity and model size comparison.

Model	Train. Params	Total Params	Size (MB)	MACs (G)	FLOPs (G)	GFLOPs/s
RawNet2	12,838,402	32,737,540	48.97	4.538	9.077	4.538
AASIST	297,866	1,312,218	1.14	4.754	9.508	4.754
AASIST3	1,451,311	6,004,919	5.54	4.827	9.653	4.827
Rawformer	345,132	2,234,021	1.32	6.186	12.371	3.093
FAN-AASIST	136,116	136,116	0.52	4.548	9.095	—

6.3 Memory Usage and Training Time

Table 4: Memory usage and wall-clock training time per model (5 epochs).

Model	Input (MB)	Fwd/Bwd (MB)	Params (MB)	Total (MB)	Train Time
RawNet2	0.13	87.69	51.35	139.17	29 min 12 sec
AASIST	0.13	434.61	1.18	435.92	55 min 32 sec
AASIST3	0.13	434.88	0.85	435.85	1 hr 43 min
Rawformer	0.13	322.39	0.69	323.21	1 hr 10 min
FAN-AASIST	—	—	0.52	—	—

AASIST and AASIST3 have surprisingly large forward/backward memory footprints ($\sim$435 MB) relative to their parameter counts ($\sim$1 MB), because graph attention operations materialise large attention matrices during the backward pass. RawNet2, despite having 50$\times$ more parameters, has the lowest activation memory (87.69 MB). Rawformer strikes the best balance: 323 MB activation memory, 1.32 MB parameters, and the best detection performance.

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6.4 Training Dynamics

Training and validation loss curves over 5 epochs reveal important differences. RawNet2 converges rapidly in training loss (0.498 → 0.00033) but shows signs of overfitting given its large capacity. AASIST and AASIST3 exhibit slower but stable convergence. Rawformer achieves the lowest validation loss (0.01567 at epoch 5) and the smoothest curve, suggesting it generalises best under this configuration.

AASIST shows an anomalous spike in training loss at epoch 4 (0.62806), likely attributable to a learning rate scheduling interaction or a difficult batch. This spike does not appear in the validation curve, suggesting it is a training-phase artifact.

7. Analysis and Discussion

7.1 Why Does Rawformer Outperform Graph-Based Models?

Rawformer's superior performance can be attributed to three factors:

- **Global attention:** Transformer self-attention allows every token to attend to every other, enabling detection of long-range spectro-temporal dependencies that graph attention—limited to local neighbourhood aggregation—may miss.
- **Longer input context:** Rawformer processes 4 seconds vs. 2 seconds, providing more temporal context for detecting artifacts in longer synthetic utterances.
- **Positional encoding:** 2D positional encoding explicitly preserves the spatial structure of the feature map before flattening, giving the Transformer awareness of where each token originates.

7.2 Why Did FAN-AASIST Underperform?

FAN-AASIST achieves a testing accuracy of 75.4% and an EER of 0.2515—worse than all other models. Several factors likely contribute:

- **Architectural mismatch:** FAN layers are designed for sequential data with natural periodic structure. The graph node embeddings in AASIST do not have an inherent sequence order, so the periodicity assumption may not hold.
- **Insufficient training:** With only 5 epochs, the learnable frequency matrices W_p in FAN may not have converged to meaningful values.
- **Scale mismatch:** FAN-AASIST has 136K parameters—roughly half of AASIST—which may be insufficient to model complex cross-modal spoofing artifacts.
- **Future direction:** Applying FAN layers to the temporal sequence after flattening (similar to Rawformer's Transformer) rather than within graph attention may be a better architectural choice.

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7.3 Cross-Model Architectural Comparison

Table 5: Architectural design principle comparison.

Criterion	RawNet2	AASIST	AASIST3	Rawformer
Sequence Model	GRU	Graph Att.	KAN Graph Att.	Transformer
Feature Space	1D temporal	2D spec.-temp.	2D spec.-temp.	2D spec.-temp.
Model Size	Very large	Very small	Medium	Small
Strength	End-to-end	Compactness	Best graph EER	All metrics
Train Time	29 min	55 min	1 hr 43 min	1 hr 10 min
Input Length	2 sec	2 sec	2 sec	4 sec

8. Open Research Problems

Generalisation to Unseen Attacks

Models trained on ASVspoof 2019 perform significantly worse on newer attack systems (A18–A32 in ASVspoof 2021). Building models that generalise across attack types remains the central open problem.

Cross-Dataset Transfer

Countermeasures trained on one corpus (e.g., ASVspoof) typically degrade on in-the-wild audio. Domain adaptation and multi-dataset training are active research areas.

Nested Learning (Proposed Future Direction)

A potential next step is training a meta-model that orchestrates multiple specialised countermeasures, each targeting a different spoofing attack class—analogous to mixture-of-experts architectures.

FAN Layer Optimisation

FAN-AASIST's poor performance opens the question of how periodic decomposition can be better integrated into graph-based systems. Applying FAN after flattening (as a sequence model rather than within graph attention) is a promising direction.

Lightweight Deployment

AASIST at 298K parameters demonstrates that large models are not required. Further pruning, quantisation, and knowledge distillation could enable on-device deployment in voice assistants.

9. Conclusion

This project traced the architectural evolution of audio deepfake detection from end-to-end 1D convolutional pipelines (RawNet2) through graph-based spectro-temporal models (AASIST, AASIST3) to transformer-based encoders (Rawformer), and proposed a new variant (FAN-AASIST) that attempts to leverage Fourier decomposition for periodic artifact detection.

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The key finding is that **Rawformer, with global self-attention and a longer input context, achieves the best detection performance** (EER = 0.1427, AUC = 0.9426) among all evaluated models, despite being 37× smaller than RawNet2. AASIST remains the most parameter-efficient model at 298K. KAN layers in AASIST3 provide a measurable improvement in EER (0.1821 → 0.164) at the cost of a 4.9× parameter increase.

FAN-AASIST, while not yet competitive, represents a novel architectural hypothesis: that periodic decomposition is a natural fit for detecting deepfake artifacts. The failure of the current integration points to a mismatch between FAN's sequential inductive bias and the graph-node structure of AASIST, rather than a fundamental flaw in the hypothesis itself. Future work will explore nested learning strategies, cross-dataset generalisation, and more principled integration of Fourier-based layers into anti-spoofing pipelines.

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